

Name: _____ (please print)

Signature: _____

ECE 2202 – Exam 2

October 11, 2025

**Keep this exam closed until you
are told to begin.**

1. This exam is closed book, closed notes. A formula sheet is provided in the back. Write your ID number at the top of every page.
2. Show all work on these pages. Show all work necessary to complete the problem. A solution without the appropriate work shown will receive no credit. A solution that is not given in a reasonable order will lose credit. Clearly indicate your answer (for example by enclosing it in a box).
3. Show all units in solutions, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 100 minutes to work on this exam.

ID number: _____

GRADE

1. _____/20

2. _____/20

3. _____/20

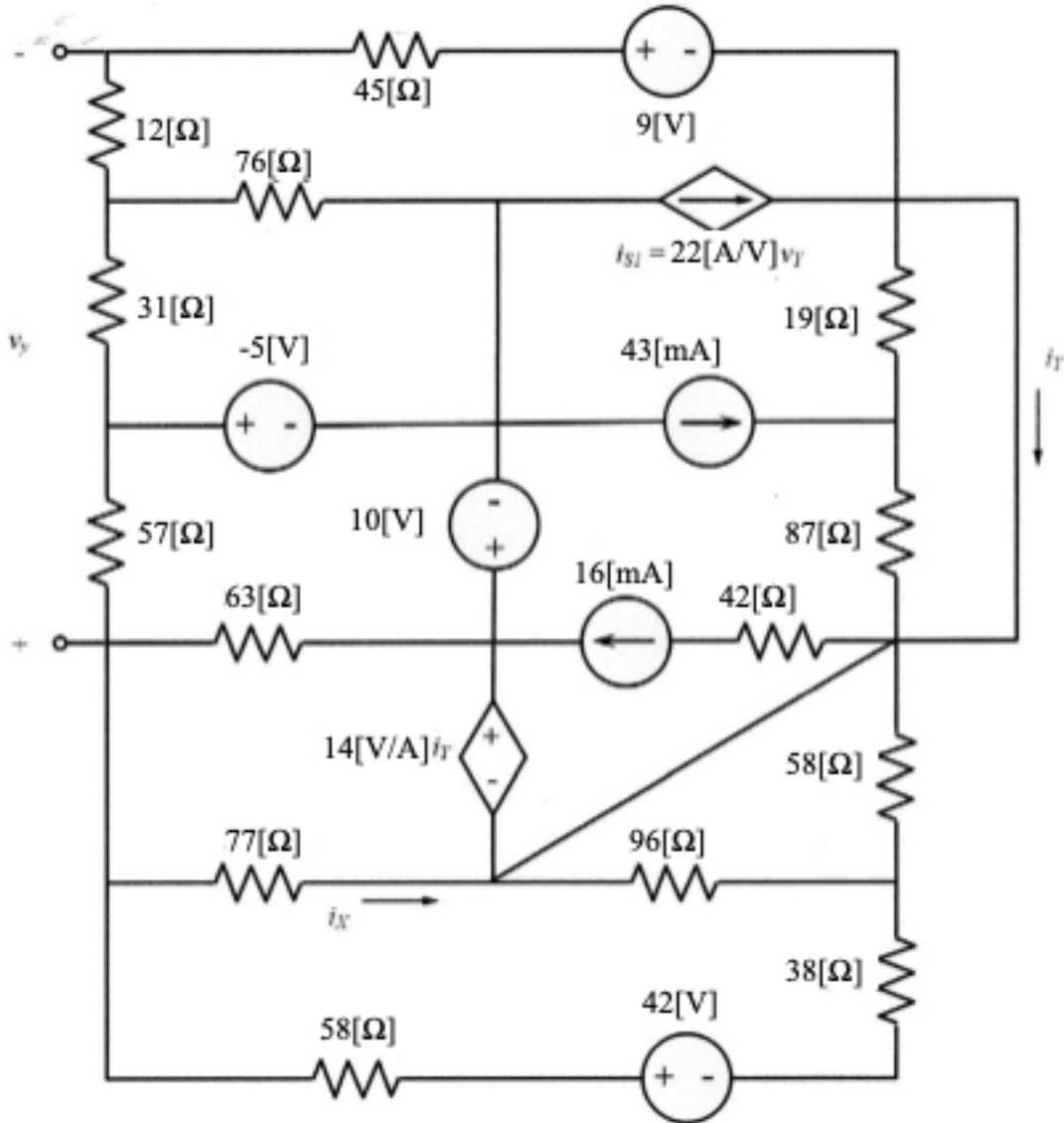
4. _____/20

5. _____/20

Total = 100

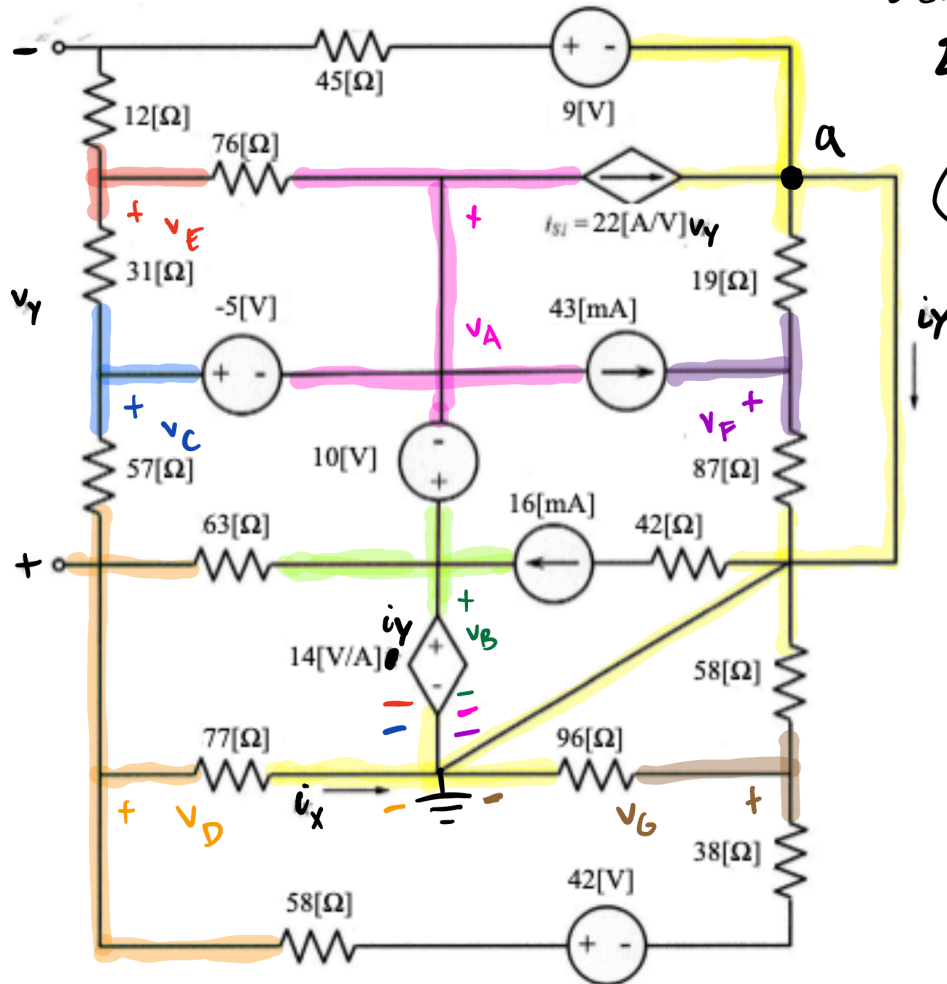
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1. {20 Points} For the following circuit, use the **node voltage method** to write a complete set of equations that could be used to solve the circuit. Do not simplify the circuit. Do not attempt to solve the equations. Label your node voltages clearly.



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8 essential nodes,
2 dependent
source
(9 equations)

(+1) labeling
(+1) organization

next page

$$1: v_A = -10 [V] + 14 [V/A] i_Y \quad (+1)$$

$$2: v_B = 14 [V/A] i_Y \quad (+1)$$

$$3: v_C = -5 [V] - 10 [V] + 14 [V/A] i_Y \quad (+1)$$

$$4: \frac{v_D - v_B}{63 [\Omega]} + \frac{v_D}{77 [\Omega]} + \frac{v_D - v_C}{57 [\Omega]} + \frac{v_D - 42 [V] - v_G}{58 [\Omega] + 38 [\Omega]} = 0 \quad (+3)$$

$$5: \frac{v_E - 9 [V]}{12 [\Omega] + 45 [\Omega]} + \frac{v_E - v_A}{76 [\Omega]} + \frac{v_E - v_C}{31 [\Omega]} = 0 \quad (+2)$$

$$6: -0.043 [A] + \frac{v_F}{19 [\Omega]} + \frac{v_F}{87 [\Omega]} = 0 \quad (+2)$$

$$7: \frac{v_G}{58 [\Omega]} + \frac{v_G}{96 [\Omega]} + \frac{v_G + 42 [V] - v_D}{38 [\Omega] + 58 [\Omega]} = 0 \quad (+3)$$

$$\textcircled{KVL} \quad v_Y: v_Y - \left(\frac{v_E - 9 [V]}{12 [\Omega] + 45 [\Omega]} \right) \cdot 12 [\Omega] + v_E - v_D = 0 \quad (+3)$$

$$i_Y: i_Y - 22 [A/V] v_Y - \left(\frac{v_E - 9 [V]}{12 [\Omega] + 45 [\Omega]} \right) - \frac{v_F}{19 [\Omega]} = 0 \quad (+2)$$

(@a)

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Room for extra work

ID number: _____

2. {20 Points} A Device shown in Figure 1 can be modeled as a voltage source in series with a resistor (Thevenin equivalent model).

a) Use Figure 2 data to find the Thevenin equivalent model of the Device. Draw and clearly label the model of the Device.

b) The Device is inserted into a circuit as shown in Figure 3. Find a Norton equivalent model of the Figure 3 circuit as seen by the 3.2[V] voltage source.

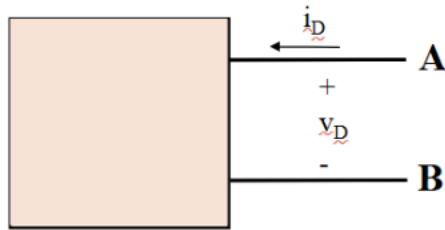


Figure 1

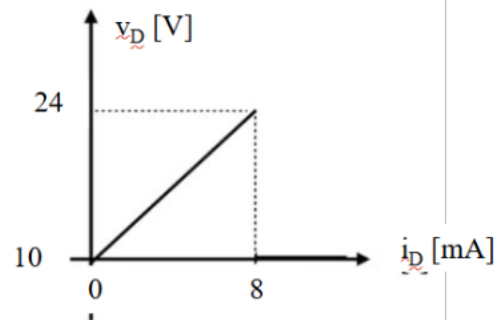


Figure 2

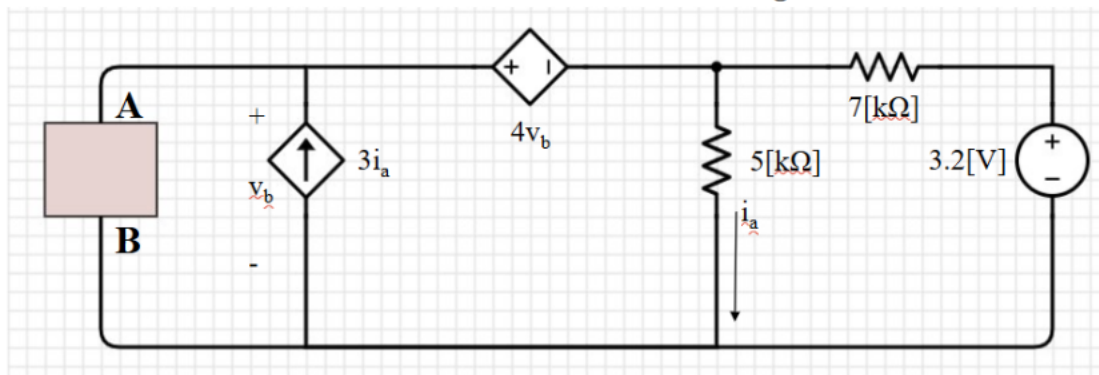
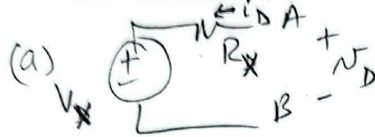


Figure 3

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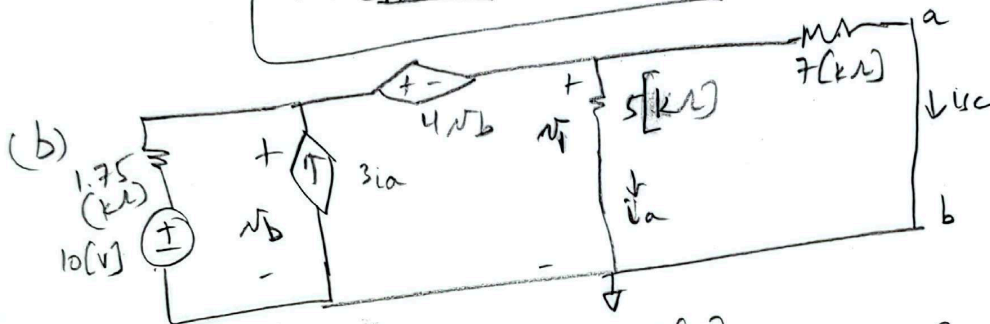
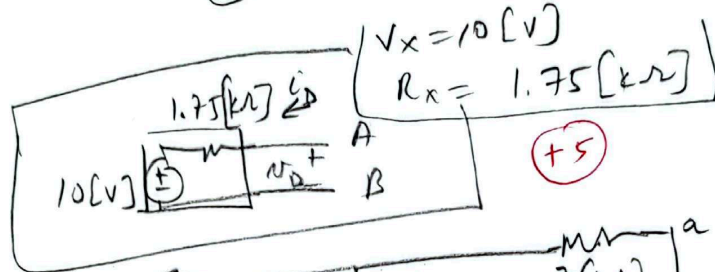
Room for extra work



$$-V_x - i_D R_x + V_b = 0$$

$$\textcircled{1} -V_x - 0(R_x) + 10[V] = 0$$

$$\textcircled{2} -V_x - 8[mA] R_x + 24[V] = 0$$



find i_{sc} :

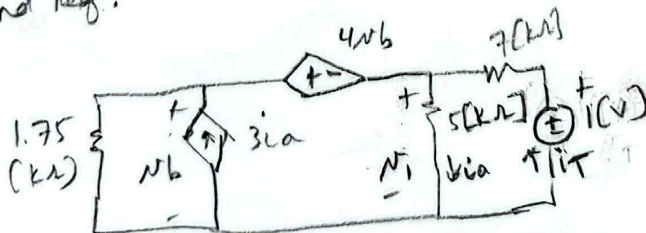
$$\frac{V_1}{5[k\Omega]} + \frac{V_1}{7[k\Omega]} + \frac{V_b - 10[V]}{1.75[k\Omega]} - 3i_a = 0$$

$$V_b - V_1 = 4V_b$$

$$V_1 = -12.766[V], i_a = -0.0026[mA]$$

$$V_b = 4.2553[V]$$

$$i_{sc} = \frac{V_1}{7[k\Omega]} = -1.8[mA]$$

find R_{eq} :

$$\frac{V_b}{1.75[k\Omega]} - 3i_a + \frac{V_1}{5[k\Omega]} + \frac{V_1 - 10[V]}{7[k\Omega]} = 0$$

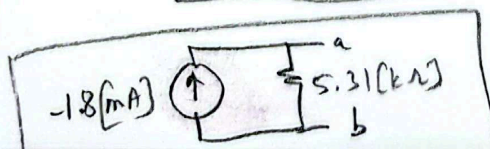
$$V_b - V_1 = 4V_b$$

$$V_1 = -0.3191[V], i_a = -0.64\mu A$$

$$V_b = 0.1064[V]$$

$$i_T = -\left(\frac{V_1 - 10[V]}{7[k\Omega]}\right) = 1.885[mA]$$

$$R_{eq} = 5.31[k\Omega]$$



$\textcircled{+3}$ overall
 units,
 clarity,
 attempt.

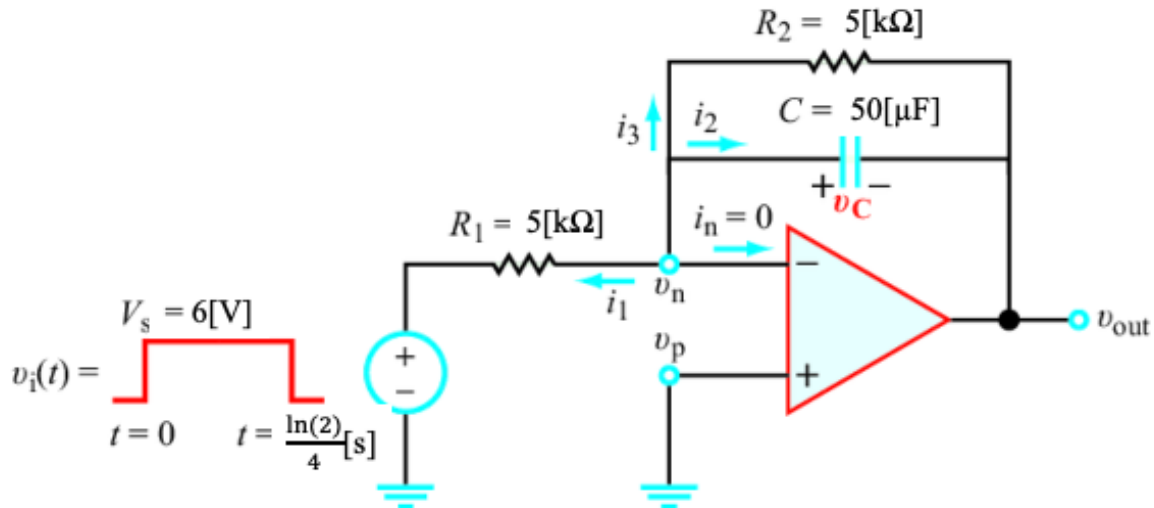
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3. {20 Points} Refer to the figure below.

The op-amp circuit below is subjected to an input pulse of amplitude $V_s = 6[V]$ and a duration $T_o = \frac{\ln(2)}{4}[s]$. Determine and plot the output voltage v_{out} for $t \geq 0$. The capacitor is charged before $t = 0$ and $v_c(0^-) = -3[V]$.

ODE: $\frac{dx}{dt} + ax = b$, General solution: $x(t) = x(0)e^{-at} + \frac{b}{a}(1 - e^{-at})$

ODE: $\frac{dx}{dt} + ax = 0$, General solution: $x(t) = x(0)e^{-at}$



① first time segment: $0 < t \leq \ln(2)/4 [s]$

(node v_n) $i_1 + i_2 + i_3 + i_n = 0$ (+2)

$$\frac{v_n - V_s}{R_1} + C \frac{d(v_n - v_{out,1})}{dt} + \frac{v_n - v_{out,1}}{R_2} = 0$$

$$-\frac{V_s}{R_1} - C \frac{dv_{out,1}}{dt} - \frac{v_{out,1}}{R_2} = 0 \Rightarrow$$

$$\boxed{\text{Eqn. 1}} \quad -\frac{V_s}{R_1 C} = \frac{dv_{out,1}}{dt} + \frac{1}{R_2 C} v_{out,1} \quad (+2)$$

first order diff. eqn. form

$$b = \frac{dx}{dt} + ax$$

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$$b = -\frac{V_s}{R_1 C}, a = \frac{1}{R_2 C} \Rightarrow \text{general solution: } v_{out,1}(t) = \underbrace{v_{out,1}(0)}_{\uparrow} e^{-at} + \frac{b}{a} (1 - e^{-at})$$

$$b = -24, a = 4$$

still unknown

$$v_{out,1}(0) = -v_c(0) = -(-3 \text{ [V]}) = 3 \text{ [V]} \quad (+2)$$

$$v_{out,1}(t) = 3e^{-4t} - 6(1 - e^{-4t}) \text{ [V]}$$

$$v_{out,1}(t) = -6 + 9e^{-4t} \text{ [V] for } 0 < t \leq \frac{\ln(2)}{4} \text{ [s]} \quad (+3)$$

② second time segment: $t > \ln(2)/4$ [s]

we know for $v_{out,2}(t)$:

a) $v_i = V_s = 0$ [V]

b) t should be defined as $t - \frac{\ln(2)}{4}$

c) $v_{out,2}(\ln(2)/4) \neq 0$

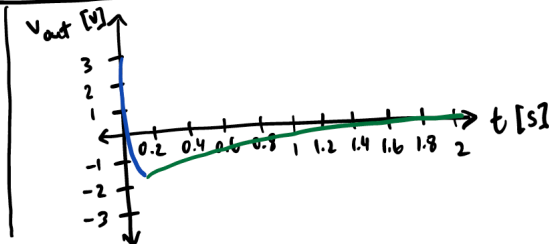
back to Eqn 1: $-\frac{V_s}{R_1 C} = \frac{dv_{out,2}}{dt} + \frac{1}{R_2 C} v_{out,2} \Rightarrow 0 = \frac{dv_{out,2}}{dt} + \frac{1}{R_2 C} v_{out,2} \quad (+2)$

first order diff. eqn. form

general solution: $v_{out,2}(t) = v_{out,2}(0) e^{-at}$ initial condition: $\frac{\ln(2)}{4}$

$$v_{out,2}\left(\frac{\ln(2)}{4}\right) = v_{out,1}\left(\frac{\ln(2)}{4}\right) = -6 + 9e^{-4(\ln(2)/4)} = -1.5 \text{ [V]} \quad (+2)$$

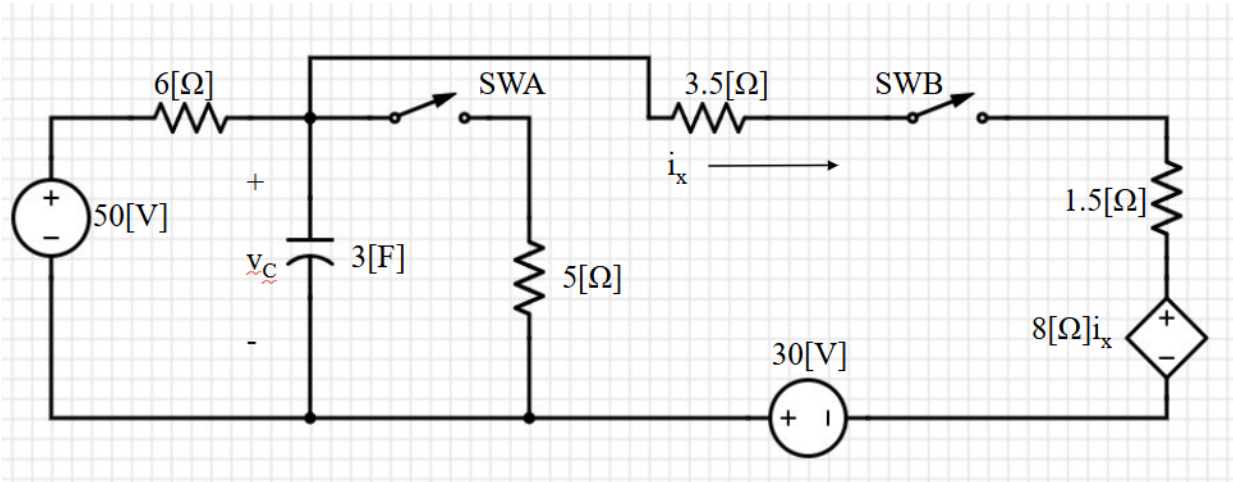
$$v_{out,2}(t) = -1.5 e^{-4(t - \frac{\ln(2)}{4})} \text{ [V] for } t > \frac{\ln(2)}{4} \quad (+3)$$



(+2) correct shape

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4. {20 Points} Switch SWA was closed and switch SWB was open for a long time. At $t=0$, switch SWA opened. Then at $2.1[s]$, switch SWB closed. Find the voltage $v_c(t)$ at $t=3[s]$.

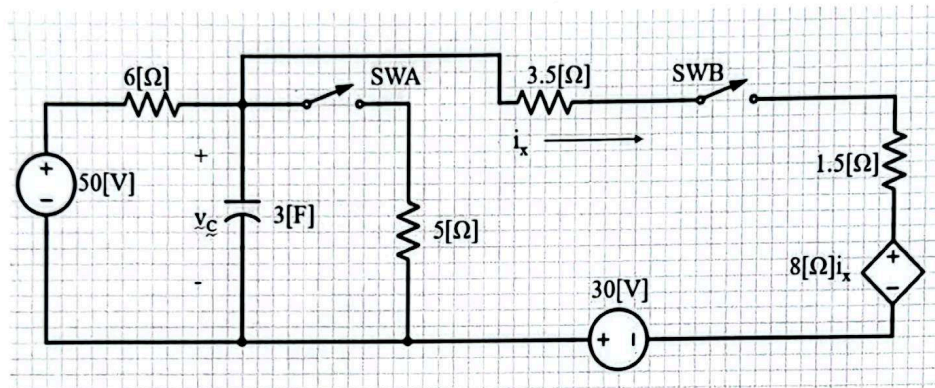


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4. {20 Points} Switch SWA was closed and switch SWB was open for a long time. At $t=0$, switch SWA opened. Then at $2.1[s]$, switch SWB closed. Find the voltage $v_c(t)$ at $t=3[s]$.



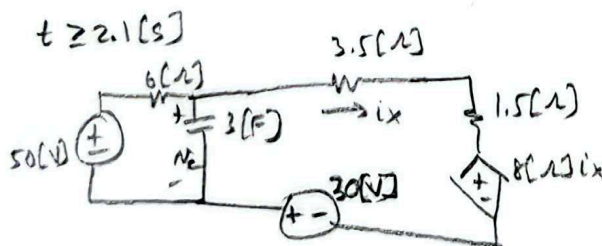
$t < 0$:

$$v_c(0^-) = 50[V] \left(\frac{5}{5+6} \right) = 22.727[V] \quad (+4)$$

$0 \leq t < 2.1[s]$

$$v_c(t) = 50[V] + (22.727[V] - 50[V]) e^{-t/(18[s])}$$

$$v_c(2.1[s]) = 25.731[V] \quad (+4)$$



at $t \rightarrow \infty$:

$$-50[V] + i_x(11[\Omega]) + 8i_x - 30[V] = 0$$

$$i_x = 4.211[A]$$

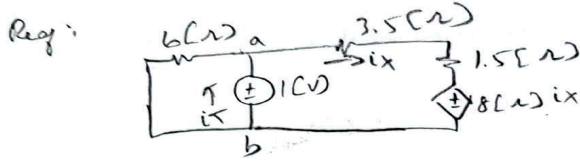
$$\frac{50[V] - v_c(\infty)}{6[\Omega]} = 4.211[A]$$

$$v_c(\infty) = 24.737[V] \quad (+4)$$

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Room for extra work



$$\frac{1[V]}{6[\Omega]} - i_T + \frac{1[V] - 8[\Omega] i_x}{5[\Omega]} = 0$$

$$i_x = \frac{1[V] - 8[\Omega] i_x}{5[\Omega]}$$

$$i_x = 0.0769[A]$$

$$i_T = 0.2436[A]$$

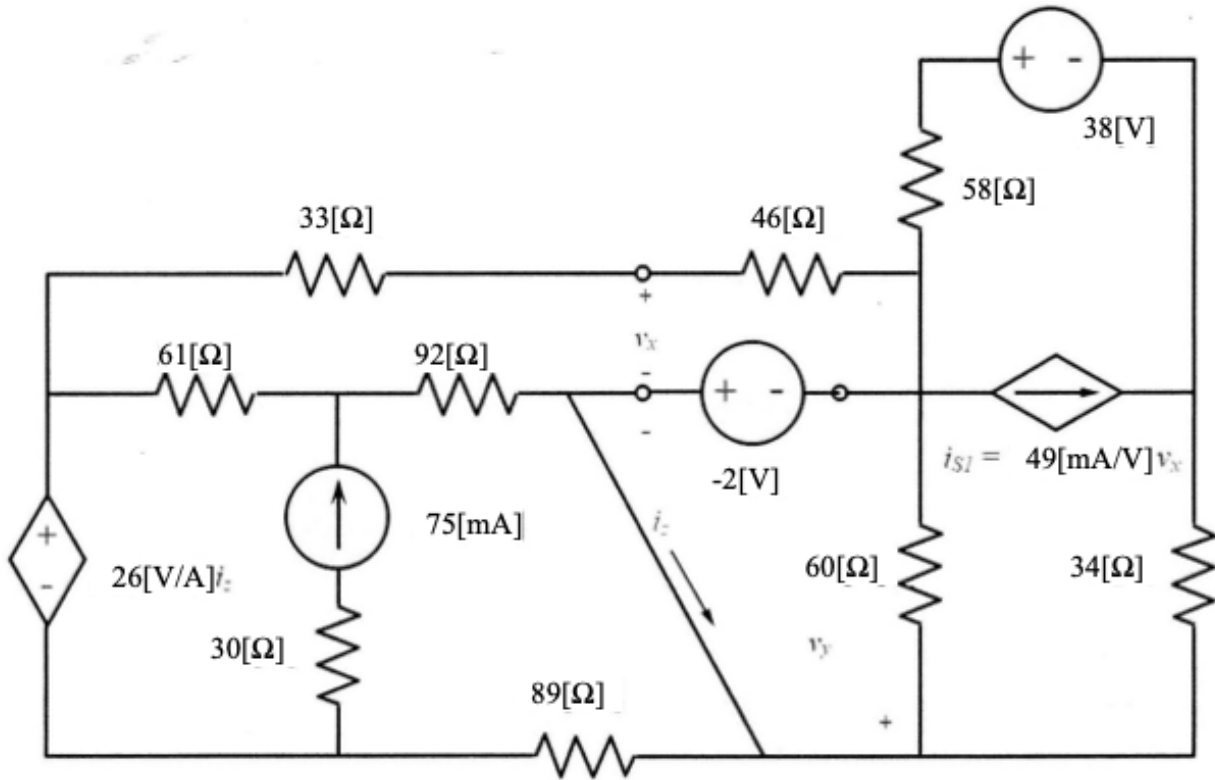
$$R_{eq} = 4.105[\Omega]$$

$$\tau = R_{eq} (3[F]) = 12.316[s]$$

$$v_c(t) = 24.737[V] + (25.731[V] - 24.737[V]) e^{-(t-2.1[s])/12.316[s]}$$

$$v_c(3[s]) = 25.661[V]$$

5. {20 Points} For the circuit below, write a complete set of **mesh current equations** that could be used to solve the circuit. Do not simplify the circuit in any way. Do not attempt to solve the equations. Label your mesh currents clearly.



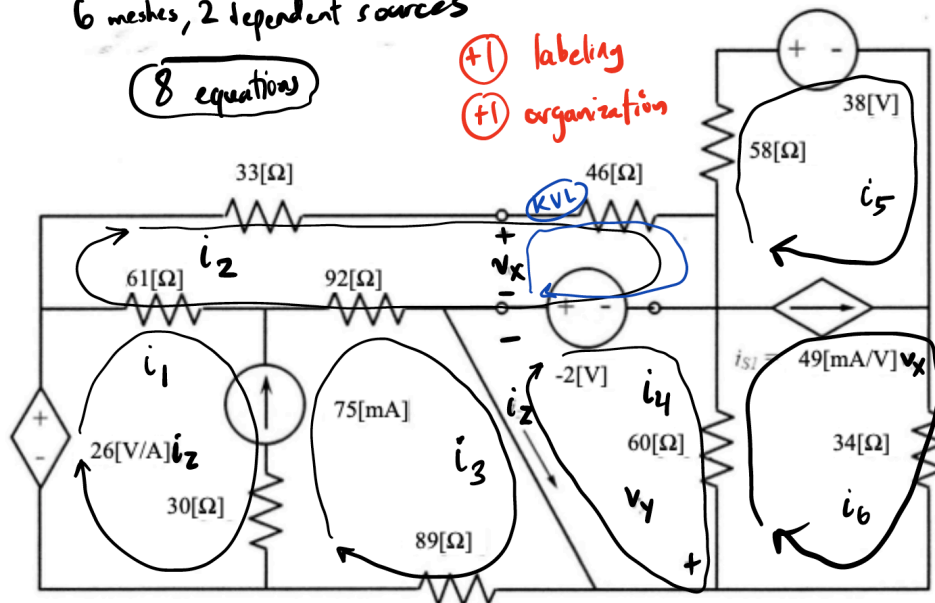
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6 meshes, 2 dependent sources

8 equations

+1 labeling

+1 organization

super mesh ($i_1 + i_3$):

$$1: i_3 - i_1 = 75 \text{ mA} \quad (+2)$$

$$2: -26 \text{ [V/A]} i_2 + (i_1 - i_2) 61 \text{ [}\Omega\text{]} + (i_3 - i_2) \cdot 92 \text{ [}\Omega\text{]} + i_3 \cdot 89 \text{ [}\Omega\text{]} = 0 \quad (+3)$$

supermesh ($i_5 + i_6$):

$$3: i_6 - i_5 = 49 \text{ [mA/V]} v_x \quad (+2)$$

$$4: i_5 \cdot 58 \text{ [}\Omega\text{]} + 38 \text{ [V]} + i_6 \cdot 34 \text{ [}\Omega\text{]} + (i_6 - i_4) \cdot 60 \text{ [}\Omega\text{]} = 0 \quad (+3)$$

$$5: i_2 \rightarrow (i_2 - i_1) \cdot 61 \text{ [}\Omega\text{]} + i_2 \cdot 33 \text{ [}\Omega\text{]} + i_2 \cdot 46 \text{ [}\Omega\text{]} - (-2) \text{ [V]} + (i_2 - i_3) \cdot 92 \text{ [}\Omega\text{]} = 0 \quad (+2)$$

$$6: i_4 \rightarrow -2 \text{ [V]} + (i_4 - i_6) \cdot 60 \text{ [}\Omega\text{]} = 0 \quad (+2)$$

next page

$$7: v_x \rightarrow \text{KVL} \quad -v_x + i_2 \cdot 46[\Omega] - (-2)[V] = 0 \quad (+2)$$

$$8: i_2 \rightarrow i_3 - i_4 = i_2 \quad (+2)$$